



CMPT 310

Project:Calgary Residential Development ROI Classifier

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Real Estate Investment Classification

The Classifier

We have designed a machine learning model to classify the townhouse development investment potential of a parcel of land in Calgary.

Model Inputs

- Address
- Community
- Purchase Price
- Number of Units

Model Output

An ordinal classification representing the overall ROI potential:

- **Low Potential**
- **Medium Potential**
- **High Potential**

AI Methods

Features

- Initial Assessment
- Neighbourhood Income
- Distance to Downtown
- Transit Stop Distance
- Cost Per Unit
- Distance to Park
- Nearest Public School

Random Forest

Decision tree ensemble classifier built from scratch.
Generates robust predictions by aggregating multiple individual decision pathways.

Ordinal Logistic Regression

Custom statistical model designed for ordered outputs.
Preserves natural relationship of investment categories: Low, Medium, and High ROI.

Ensemble Voter

Utilizes a soft voting framework for final decisions.
Pools predictions from both constituent models to output a balanced classification.

AI Pipeline

1 • Sources

Permits, assessments, income and amenities



2 • Prepare

Clean addresses, group permits and match records



3 • Engineer

Build decision tree, random forest, and ordinal logistic regression, integrate into ensemble voter

4 • Evaluate

Fold-local training; held-out real projects



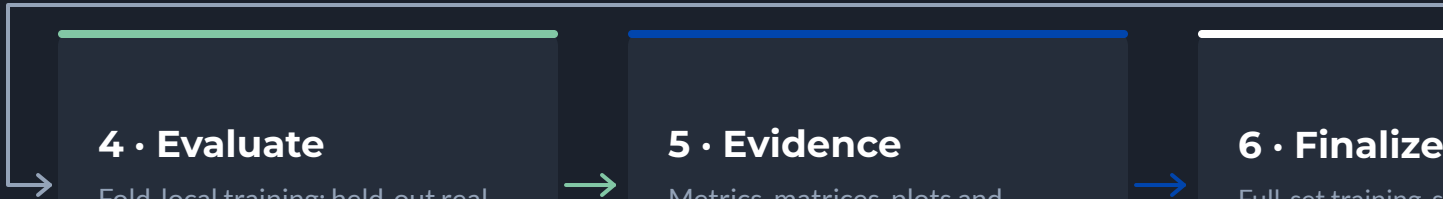
5 • Evidence

Metrics, matrices, plots and hyperparameter tuning



6 • Finalize

Full-set training, store model and run from the system interface



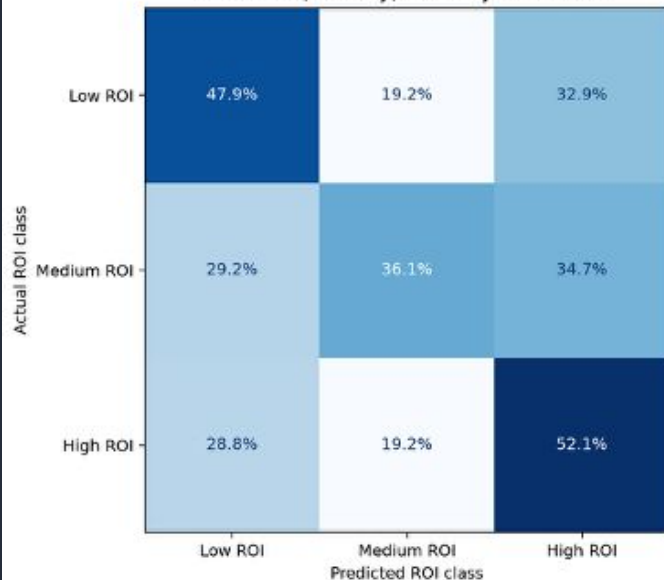
Evaluation and Interpretation

Performance Metrics

Evaluation with 5-fold stratified cross validation.

Model	Training	Training Accuracy	Accuracy
Majority baseline	Real only	33.7%	32.6% $\pm$ 0.9%
Ensemble	Real only	61.1%	45.4% $\pm$ 4.0%
Ensemble	+1,200/fold for RF Real only for OLR	49.1%	40.8% $\pm$ 5.0%
Ensemble	Real only for RF +1,200/fold for OLR	49.8%	45.8% $\pm$ 8.3%
Ensemble	+1,200/fold for RF +1,200/fold for OLR	52.7%	41.7% $\pm$ 6.9%

Ensemble (real only): recall by actual class





Evaluation and Interpretation

Real Data Only Ensemble

- **Tighter spread** across validation folds, indicates higher reliability.
- **Better performance** for correctly identifying and high value opportunities.

Synthetic Regression & Real Random Forest Ensemble

- **Highest average score** of **46.1%** achieved via ordinal logistic regression.
- **Best fold high score** of **50%** reached in peak validation runs.
- **Widest spread** across folds, presenting higher variance and overall lower stability.



Limitations

01

Training Data

Highly limited amount of real-world training data available. Only 218 real data points.

02

Feature Quality

Low correlation features make pattern recognition difficult.

03

Market Nature

Inherent unpredictability of the real estate market.



Risk Mitigation

01

Synthetic Data

To aid the issue of limited data, we generate synthetic data. Training the models with this data lead to an overall increase in average score.

02

Distribution Fitting

Training with the synthetic data comes with the risk of fitting the model to the underlying distribution used to generate the data.

Demonstration

End

